

- Bringing offline data online is a challenge.
 - Managing regulatory compliance, establishing data security, and data privacy are all challenging tasks.
- A poor data management strategy may lead to an enormous amount of data in a completely unmanageable format.

Next-generation data capabilities

To manage the huge and different types of data that enterprises collect and process, they must build the following data capabilities.

- **Data processing techniques:** Unstructured data processing covers entity extraction, concept extraction, sentiment analysis, NLP, ontology, etc. To automate data extraction, machine learning techniques must be leveraged.
- **Agile data delivery:** Agile development methodology can be used to reduce the overall cost of data delivery.
- **Digital 360-degree view of the customer and products:** Customers interact with enterprises through social media channels. The ability to create a single view of a customer across internal and external data is critical in a digital world. This applies to products too.
- **On-demand data services:** On-demand data services should be available for both online and offline use. The ability to query in a federated fashion for both online and offline access is important.
- **Unstructured metadata:** Data is being thrown into enterprise systems at a rate that doesn't allow smaller businesses to look at data structures and extract metadata. Automated metadata extraction based on ontology is critical.
- **Real-time analytics:** Multi-channel applications and decision management systems need to be used to capture interactions for digital processes in real-time scenarios.

- **Data archival:** Compliance and performance requirements drive the need for archival of both structured and unstructured data.

Principles of data architecture

Data architecture principles are a structured set of ideas that collectively define and guide the development of a solution architecture — from values to design and implementation, harmonising decision-making across an organisation. A few principles are listed below.

- **Data is an asset:** Data is an asset that has a specific and measurable value for the enterprise.
- **Data is shared:** Users have access to the data necessary to perform their duties; therefore, data is shared across enterprise functions and organisations.
- **Data trustee:** Each data element has a trustee accountable for data quality.
- **Common vocabulary and data definitions:** Data definition is consistent throughout the enterprise, and the definitions are understandable and available to all users.
- **Data security:** Data is protected from unauthorised use and disclosure.
- **Data privacy:** Data privacy and protection are paramount throughout the life cycle of a Big Data project. All data sharing will conform to the relevant regulatory and business requirements.
- **Data integrity and the transparency of processes:** Each party to a data analytics project must be aware of and abide by its responsibilities regarding the provision of source data, and the obligation to establish and maintain adequate controls over the use of personal or other sensitive data.
- **Fit for purpose tools:** The next-generation information ecosystem needs to have fit for purpose tools, as one technology will not meet the

requirements of all the workloads and processing techniques.

- **Data governance:** As data resides around many repositories, integrated metadata lineage and PII (personally identifiable information) protection are important.

The key technology trends that are driving enterprise data management are depicted in Figure 1. These trends help to establish more dynamic, secure and user-friendly data management practices across the enterprise.

Key components of data management

Data collection: This is the process of collecting data from various sources (internal and external) to provide the insights required to solve various business problems. The main data sources today are social media, mobile devices, cloud computing, real-time sensor data, AI/ML, cybersecurity and predictive analytics.

As an example, for a trained AI model to carry out sentiment analysis, the collected data should include a large number of reviews, comments, and social media posts. Web scraping is the automated method of extracting data from various websites.

Notable open source data collecting tools are ODK (Open Data Kit), KoboToolbox, Jotform, and Fulcrum.

Data storage: This helps to determine where and how the data will be stored as an enterprise data repository. It covers both structured and unstructured data sources, which includes databases, data warehouses and data lakes.

Key open source players in this field are Hadoop, LakeFS, Cassandra, and Neo4J. These tools provide scalability, robustness and performance in managing large data and analysing large data sets in various applications.

Data processing: Data is pre-processed and undergoes cleansing, normalisation, and tokenisation to derive insights from it that support business processes.